

Gayatri Kharche

San Diego, CA | +1 858-518-9520 | gkharache@ucsd.edu | gayatrikharche | [LinkedIn](#) | [Portfolio](#) | [Github](#)

EDUCATION

UC San Diego

Master of Science in Machine Learning and Data Science

San Diego, CA

Sep. 2023 - Mar. 2025

University of Pune

Bachelor of Engineering in Electronics and Telecommunication (GPA: 4.0/4.0)

Pune, India

Aug. 2019 - Jun. 2023

WORK EXPERIENCE

Machine Learning Engineer

San Diego, CA

San Diego Supercomputer Center

Sep. 2024 - Present

- Spearheaded the development of an LLM-powered chat system with an integrated AI agent for real-time data fetching from Thanos (NRP), reducing manual data retrieval time by 80%.
- Deployed Llama3 using LiteLLM proxy, optimizing model performance and enhancing conversational AI capabilities.
- Integrated JavaScript-based interactive agents to generate plots from Prometheus.
- Optimized real-time inference pipelines for LLM, ensuring low-latency and maximizing model execution efficiency.

Machine Learning Intern

San Diego, CA

San Diego Supercomputer Center

Jul. 2024 – Sep. 2024

- Benchmarked ML models, including transformers and ResNet50, on OSPool, processing 150MB–150GB datasets over 685 GPU hours to optimize model execution and resource utilization.
- Diagnosed GPU cluster job efficiency, revealing 47% compute time, 7% file transfers, and 44% failed transfers, exposing bottlenecks in job completion.
- Devised an ensemble technique, cutting file transfer time by 50% by minimizing job stalling and eviction.
- Evaluated job performance across multiple sites, pinpointing optimizations for distributed ML workflows and enhancing execution efficiency.

Software Engineer Intern

Pune, India

Hewlett Packard Enterprise

Feb. 2023 – Aug. 2023

- Engineered scalable virtualized environments via VMware and AWS, accelerating workload processing by 30% and optimizing resource allocation by 15%.
- Orchestrated DNS, NTP, and vCenter configurations, fortifying reliability and compressing deployment time by 20%.
- Mastered HPE GreenLake, hardware systems, cloud infrastructure, networking protocols, VMware, AWS, and Linux, cultivating a strong technical foundation for enterprise-level IT solutions.

Data Analyst

Pune, India

Prakalp Infrabuild Pvt. Ltd.

Aug. 2022 – Jan. 2023

- Synthesized project data into actionable insights by leveraging advanced SQL queries and data visualization tools (Tableau), resulting in a 10% reduction in project delivery time and enhanced cost management.
- Designed and deployed ETL pipelines to automate data flow enhancing reporting accuracy and business intelligence.

PROJECTS

FineShot: Fine-grained Zero-shot Object Detection using VLMs

Jan. 2025 - Mar. 2025

- Engineered various Zero-shot object detection models using OWLv2, YOLO-World, and VLMs like Qwen2.5-VL and CLIP, achieving detection for fine-grained labels.
- Optimized YOLOv11 object detection model by integrating Vision Language Models (VLMs), boosting the Mean Average Precision (MAP) score by 15% without requiring additional retraining cycles or resources.

Hallucination Detection in Llama 2B

Oct. 2024 - Dec. 2024

- Achieved 68% accuracy on hallucination detection models with transformer architectures (DistilBERT, ALBERT, BERT, RoBERTa) on the Fever dataset, improving overall model performance.

SKILLS

Languages and Tools: Python, SQL, C++, Bash Scripting, Linux, HTML/CSS, Matlab, NoSQL, JavaScript, Java, R, Git, Jira

Machine Learning & AI: PyTorch, TensorFlow, RAGs, LangChain, LLM Finetuning, Transformers, LLM Agents, GenAI, Vector Databases, Deep Learning, Prompt Engineering, CUDA, Diffusion. vLLM, Distributed Training, Model Optimization.

Data & Computing: Apache Spark, Tableau, Hadoop, AWS, Docker, Kubernetes, MLOps, Azure AI Studio, GCP.